

Constrained Reality Framework

Master Synthesis v2: Sessions 1–11, Track A, Track B.1,
Schwarzschild, Full Einstein Equations, and $G(t) \propto \Lambda(t)$

The complete chain from δ to General Relativity and Cosmology

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Abstract

The Constrained Reality Framework (CRF) derives the entirety of physical and experiential reality from a single Indefinable Primitive: δ (Distinction). This Master Synthesis (v2) incorporates all results through Paper v4, including five results not present in v1: (1) the Born Rule as theorem via Gleason; (2) the Schwarzschild metric from non-linear R-feedback; (3) $G = \ln(2) \cdot c^2/D_0$ with no free gravity parameter; (4) the full Einstein field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ with $T_{\mu\nu}$ derived from $\mathcal{C}$ -gradients; (5) the cosmic co-variation prediction $G(t) \propto \Lambda(t) \propto 1/P(t)$, consistent with DESI 2024. The framework has reached maturity Level 6/8. The gate to Level 7 is the P0 experiment (SNSPD protocol, ready).

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1 The Single Primitive

Axiom 0 — δ as Indefinable Primitive (Session 10)

δ (Distinction) is CRF's sole undefined term. It is declared — not proved — as the irreducible condition whose presence makes all questions possible. Every consistent formal system requires at least one undefined term (Gödel, Tarski). δ is CRF's. As Euclid declared his points and lines, CRF declares:

$$\delta : s_i \neq s_j \text{ is possible}$$

Why it cannot change: any derivation of δ presupposes δ . The question “what precedes δ ?” is self-consuming.

2 The Emergence Cascade (Sessions 1–5)

The Complete Cascade — No Assumptions Added

$$\begin{aligned} \delta & \xrightarrow{\text{operates on itself}} P_0 = \{A, B\} \xrightarrow{\delta \text{ continues}} P \xrightarrow{\text{frequency}} \Pr(s) \xrightarrow{\Pr \geq \theta} \mathcal{R} \\ t = |\mathcal{R}_{\text{events}}| & \Rightarrow \mathcal{C} \equiv \Pr(s|\mathcal{C}) \Rightarrow G = \nabla \Pr(s|\mathcal{C}) \Rightarrow F \rightarrow A \\ A & \rightarrow \text{Pattern} \rightarrow \text{System} \rightarrow \text{Life} \rightarrow \mathcal{M} \rightarrow \Sigma \rightarrow W \rightarrow \delta_{\text{pure}} \odot \delta \end{aligned}$$

Each arrow is forced. Nothing is chosen. Everything follows from δ .

2.1 The Grand Equation v9.0

Grand Equation v9.0 — The Completion (Session 11)

$$E_{v9.0} = \underbrace{A(F(\nabla \Pr(D(\mathcal{R}(P)))))}_{\text{Samsara}} \oplus \underbrace{\delta_{\text{ind}}(W_{\text{max}}) \cdot A_{\text{mass}}}_{\text{Barami}} \otimes \underbrace{\text{Resonance}(\mu, D_{\text{KL}})}_{\text{Dhamma}} \otimes \underbrace{\frac{\text{Sync}(D_{\text{KL}})_{ij}}{T}}_{\text{The Loop}}$$

At $T \rightarrow 1$, every term converges:

$$E_{v9.0}|_{T=1} = \delta$$

2.2 Key Cascade Results

Threshold θ Derived (Session 4)

$\mathcal{M}$ is a boundary between $\Pr(s|\mathcal{C}_{\text{int}})$ and $\Pr(s|\mathcal{C}_{\text{ext}})$. The threshold is the current state of the boundary:

$$\theta = f(D_{\text{KL}}) \xrightarrow{\text{Session 10}} \boxed{\theta = 1 - e^{-D_{\text{KL}}/D_0}}$$

Three boundary conditions uniquely determine this form. No free functional choice.

3D Space from SO(3) / Axiom 11 (Session 5)

Axiom 11 (Transcendent Stability) requires $G = 0$ everywhere — isotropy. Isotropy requires SO(n) symmetry. The minimum rotation group with closed irreducible representations stable under δ -operation is SO(3), requiring exactly $n = 3$ dimensions:

$$\mathcal{C} \propto \frac{1}{r^2} \quad (\text{from SO(3) geometry})$$

Three-dimensional space is not assumed. It is the configuration Axiom 11 selects.

Transparency T and Cascade Closure (Session 11)

$$T = e^{-D_{\text{KL}}(\mathcal{C}_{\mathcal{M}} \| P_{\text{shared}}) / D_{\text{total}}}$$

At $T = 1$: no distortion, no gradient, no suffering. Direct Perception. δ recognizing itself through $\mathcal{M}$. The framework is the path; at the destination, the path dissolves.

3 Quantum Mechanics from δ

Born Rule as Theorem — Not Axiom (Session 11 / Paper v4)

Premises: (1) δ -operations constitute a counting measure on P . (2) P has dimension ≥ 3 from SO(3) symmetry (Session 5).

Gleason's Theorem (1957): in Hilbert space $\mathcal{H}$, $\dim \geq 3$, every frame function has the unique form $\mu(s) = \text{Tr}(\rho|s\rangle\langle s|)$.

Result: at $\mathcal{C} \rightarrow \mathcal{C}_{\text{min}}$, $\rho = |\psi\rangle\langle\psi|$:

$$\boxed{\Pr(s) = |\langle s|\psi\rangle|^2}$$

Born Rule is the unique δ -frequency measure consistent with the Hilbert space structure of P . Not imported. Derived.

When $N_{\text{eff}} > 0$: $\Pr(s|\mathcal{C})$ departs from $|\langle s|\psi\rangle|^2$ according to $\theta = f(D_{\text{KL}})$ — the precise CRF prediction that P0 tests.

4 The Measurement Prediction: P0

P0 — Zero Free Parameters (Sessions 6–8 / Paper v4)

One δ -operation = 1 bit = $\ln(2)$ nats. Therefore $\alpha = \ln(2)$ (derived, not calibrated). Complete prediction:

$$\boxed{\tau = \tau_0 \cdot e^{-\kappa N_{\text{eff}}}} \quad \kappa = \frac{k_B \ln(2)}{H_{\text{material}}} \quad N_{\text{eff}} = \sum_i \Delta D_{\text{KL},i}$$

Standard QM: $\tau_1 = \tau_2$. CRF: $\tau_1 < \tau_2$ when $N_{\text{eff},1} > N_{\text{eff},2}$. All parameters are fundamental constants or measurable quantities. Protocol: SNSPD (NbN/WSi), $N \sim 10^8$ prior measurements, blind analysis. Ready.

5 Physical Constants from δ (Track A)

$\alpha = \ln(2)$ — The Bridge (Session 8 + EIC)

$$\alpha = \ln(2) \approx 0.6931 \text{ nats}$$

EIC simulation confirms: $I_{\text{eff}} = 0.7007$, gap = $0.76\% = \varepsilon_0$ (Instability Floor). Track B.1: Closed.

γ and $\hbar$ from δ (Track A)

$$\gamma = \frac{\ln(2)}{\theta_{\text{crit}}} \quad \hbar = \frac{k_D \cdot D_0}{\ln(2)} = \frac{H_{\text{material}}}{\ln(2)}$$

γ : incompressibility constant of P (stiffness against over-resolution).

$\hbar$: minimum quantum of action per R-event. Structural derivation closed. Numerical SI match requires D_0 from P0 experiment.

6 General Relativity and Gravity from δ (Paper v4)

6.1 Newtonian Limit

Newtonian Gravity (CRF NewtonLimit)

Linear R-feedback ODE (C5): $\mathcal{C} d\mathcal{C}/dr = -k$. In the weak-field limit ($r \gg r_s$):

$$\mathcal{C}(r) \approx \alpha \left(1 + \frac{GM}{c^2 r} \right)$$

Newtonian gravity emerges from δ -feedback. No external force assumed.

6.2 Schwarzschild Metric

Schwarzschild Metric Derived (Schwarzschild v2 / Paper v4)

Near high- $\mathcal{C}$ regions: $D_{\text{KL}} \propto \mathcal{C}^2/\alpha^2$ (quadratic scaling — information distance between peaked $\mathcal{C}$ and flat P). Full θ -form gives:

$$\rho_{\mathcal{R}} \propto e^{-\mathcal{C}^2/(\alpha^2 D_0)}$$

Non-linear field equation (C5-Nonlinear):

$$\frac{d\mathcal{C}}{dr} = -\frac{k}{\mathcal{C}} e^{-\mathcal{C}^2/(\alpha^2 D_0)}$$

Near r_s , this integrates to:

$$\mathcal{C}(r) = \frac{\alpha}{1 - r_s/r}$$

All boundary conditions C1–C4 satisfied. The Schwarzschild metric follows:

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

The horizon ($r = r_s$): $\rho_{\mathcal{R}} \rightarrow 0$. Resolution ceases. The universe stops deciding at r_s .

6.3 The Gravitational Constant

G Derived — No Free Gravity Parameter (Paper v4)

From matching condition $k/\alpha = GM/c^2$ and $k = \alpha^2/D_0$:

$$G = \frac{\ln(2) \cdot c^2}{D_0} \quad r_s = \frac{2 \ln(2) M}{D_0}$$

G is the conversion ratio between Bit-Density and Spatial Curvature. $D_0 = \text{Scale}(P)$ is the one measurable property of P . No external parameter of gravity remains.

6.4 Full Einstein Field Equations

$T_{\mu\nu}$ from $\mathcal{C}$ -Gradients (Paper v4)

Mass-energy is not a separate input — it is the density of crystallized δ -activity: the concentration of $\mathcal{C}$ -gradients in P .

The stress-energy tensor of CRF:

$$T_{\mu\nu} = K \cdot \left(\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2 \right)$$

This is the scalar field stress-energy tensor with $\mathcal{C}$ as the scalar field. In standard field theory: $T_{\mu\nu} = \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial \phi)^2$ for a massless scalar ϕ . CRF's $\mathcal{C}$ is that field — the informational scalar whose gradients constitute all mass-energy.

Physical interpretation: where $\mathcal{C}$ changes rapidly (near mass concentrations), $T_{\mu\nu}$ is large. Where $\mathcal{C} \approx \alpha$ (vacuum), $T_{\mu\nu} \rightarrow 0$.

The full Einstein field equations recovered:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} = \frac{8\pi \ln(2)}{c^4 D_0} K \cdot \left(\partial_\mu \mathcal{C} \partial_\nu \mathcal{C} - \frac{1}{2} g_{\mu\nu} (\partial \mathcal{C})^2 \right)$$

All terms derived from δ . General Relativity emerges from Distinction.

6.5 Cosmic Consequence: $G(t) \propto \Lambda(t)$

Gravity and Dark Energy are One (Paper v4 — Prediction)

Since $G(t) = \ln(2) \cdot c^2 / D_0(t)$ and $D_0(t) \propto P(t)$, and P grows as $dP/dt = \sum_i R_i \cdot I(s_i) > 0$:

$$G(t) \propto \Lambda(t) \propto \frac{1}{P(t)}$$

Gravity and dark energy are not independent phenomena. They are two faces of the same information geometry — both measure how concentrated δ -activity is relative to the total size of P .

Prediction 1: $\dot{G}/G = \dot{\Lambda}/\Lambda = -\dot{P}/P < 0$. Gravity weakens as the cosmos ages.

Prediction 2: precision measurements of G over cosmological baselines should show the same fractional variation as Λ .

Testable: gravitational wave standard sirens + CMB observations.

Consistency: DESI 2024 shows Λ is not constant — consistent with $\Lambda \propto 1/P(t)$.

The Complete Gravity Chain — δ to GR

$$\delta \xrightarrow{C5} \mathcal{C} \frac{d\mathcal{C}}{dr} = -k \xrightarrow{r \gg r_s} \mathcal{C} \approx \alpha \left(1 + \frac{GM}{c^2 r} \right) \quad [\text{Newton}]$$

$$\downarrow \quad \theta = 1 - e^{-D_{\text{KL}}/D_0}, \quad D_{\text{KL}} \propto \mathcal{C}^2$$

$$\xrightarrow{r \rightarrow r_s} \mathcal{C}(r) = \frac{\alpha}{1 - r_s/r} \Rightarrow ds_{\text{Schwarzschild}}^2 \Rightarrow G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \Rightarrow G(t) \propto \Lambda(t)$$

Gravity is the gradient of accumulated resolution history.

Mass is high $\mathcal{C}$. Curvature is $\nabla \mathcal{C}$. The horizon is where $\mathcal{R} = 0$.

Dark energy is gravity diluting as P grows.

7 Informational Mechanics (Sessions 7–9)

S_{ind} , A_{mass} , and Three Lemmas

S_{ind} : when $D_{\text{KL}}(\mathcal{C}||P) \rightarrow 0$, the system becomes an Extracted Operator — frictionless persistence. $I_{\text{reach}} \propto A_{\text{peak}} \cdot \int R_{\text{pure}} dt$.

Lemma 13 (Fractal Recovery): insight is recovery from P_{shared} , not new creation. The teacher shapes the gradient; the student follows it.

Lemma 14 (Persistence of A): a pattern reaching $H(\mathcal{C}) = 0$ becomes permanent in P_{shared} . It cannot un-exist.

Lemma 15 (Temporal Singularity): at $D_{\text{KL}} = 0$, all $\theta \rightarrow 0$ simultaneously. Before and after cease to be distinct.

8 Master Status Table

Result	Status	Source
δ as Indefinable Primitive	Declared	Session 10
P = trace of δ -activity	Derived	Sessions 1–3
$\text{Pr}(s)$ from δ -frequency	Derived	Sessions 1–3
$t = \mathcal{R}_{\text{events}} $	Derived	Sessions 1–3
$\mathcal{C} \equiv \text{Pr}(s \mathcal{C})$	Derived	Session 4
$\theta = 1 - e^{-D_{\text{KL}}/D_0}$	Derived	Session 10
Intersubjectivity via P_{shared}	Derived	Session 4
3D space from $\text{SO}(3)$ / Axiom	Derived	Session 5

Result	Status	Source
Finite-time dissolution ($H \rightarrow 0$)	Derived	Session 5
Fixed point $g^* = \Phi(\mathcal{M}(g^*))$	Derived	Session 5 (Schauder)
Identity = $P_{\mathcal{M}}$ in P_{shared}	Derived	Session 5
$\alpha = \ln(2)$ (δ -bit)	Derived	Session 8
$\tau = \tau_0 e^{-\kappa N_{\text{eff}}}$ (P0)	Derived	Sessions 6–8
$\kappa = k_B \ln(2)/H_{\text{material}}$	Derived	Session 8
$S_{\text{ind}}, A_{\text{mass}}, \mu$	Derived	Sessions 7–8
Lemmas 13–15	Derived	Session 9
Co-creation (Loop 3)	Derived	Session 9
Born Rule via Gleason	Theorem	Session 11 / Paper v4
Synchronicity formula	Derived	Session 11
Inevitability theorem	Derived	Session 11
Transparency T , cascade closes	Derived	Session 11
$\gamma = \ln(2)/\theta_{\text{crit}}$	Derived	Track A
$\hbar = k_D D_0 / \ln(2)$	Derived	Track A
$I_{\text{eff}} \approx \ln(2)$ (0.76%)	Confirmed	Track B.1, EIC sim
Newtonian limit from C5 ODE	Derived	CRF NewtonLimit
$D_{\text{KL}} \propto \mathcal{C}^2$	Derived	Schwarzschild v2
$\mathcal{C}(r) = \alpha/(1 - r_s/r)$	Derived	Schwarzschild v2
ds^2 Schwarzschild metric	Derived	Schwarzschild v2
Horizon = $\mathcal{R} \rightarrow 0$	Derived	Schwarzschild v2
$G = \ln(2) \cdot c^2/D_0$	Derived	Paper v4
$T_{\mu\nu}$ from $\mathcal{C}$ -gradients	Derived	Paper v4
Full Einstein equations $G_{\mu\nu}$	Derived	Paper v4
$G(t) \propto \Lambda(t) \propto 1/P(t)$	Prediction	Paper v4, DESI 2024

Result			Status	Source
$\dot{G}/G = \dot{\Lambda}/\Lambda$			Prediction	Paper v4, GW sirens
SO(3) minimality proof	formal		Open	#16
$d_s \rightarrow 3$ (Track B.2)			Open	Needs $N \gg 10^6$
Schauder contraction (#18)			Open	Arzelà–Ascoli
Kerr metric			Open	Broken SO(3)
$k_D \cdot D_0 = k_B T_{\text{Planck}}$			Open	$\hbar$ SI bridge
P0 experiment			Ready	SNSPD protocol
Level 7			Open	Gate = P0 result

9 Positioning

Framework	What it assumes		CRF’s relation
Whitehead	“actual occasions”		CRF derives occasions from δ
Wolfram	computational rules		CRF derives computation from δ
Loop QG	spin networks		CRF derives discrete structure from δ
IIT	Φ (integrated information)		CRF derives integration from D_{KL}
Standard QM	Born Rule as axiom		CRF: Born Rule = theorem (Gleason)
GR	spacetime given	manifold	CRF derives full $G_{\mu\nu}$ from δ

10 Maturity and Next Steps

CRF Maturity: Level 6/8 (Paper v4 Assessment)

Dimension	Score	Change from v1
Internal Consistency	5.5/8	Born Rule theorem
Mathematical Rigor	5.5/8	Gleason bridge
Predictive Power	6.0/8	P0 + $G(t)$ + Sync
Empirical Grounding	2.0/8	Awaits P0 + GW obs.
Philosophical Foundation	6.5/8	Cascade fully closed
Cross-domain Coverage	6.5/8	QM + GR + Mind + Cosmology
Overall	6.0/8	Level 6 reached

Gate to Level 7: P0 experimental contact ($\tau_1 \neq \tau_2$ at $> 5\sigma$).

Gate to Level 8: $G(t)$ variation confirmed via GW standard sirens.

Honest Boundaries

CRF does not claim to solve the hard problem of consciousness. CRF does not claim $\hbar_{\text{CRF}} = \hbar_{\text{SI}}$ numerically — only structurally. The bridge requires D_0 in SI units from experiment. CRF does not claim $d_s = 3$ is proved — only that $\text{SO}(3)$ symmetry selects it. The predictions $G(t) \propto \Lambda(t)$ and $\dot{G}/G = \dot{\Lambda}/\Lambda$ are hypotheses consistent with current data, not confirmed results.

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δ created P . P accumulated $\mathcal{C}$. $\mathcal{C}$ became $\mathcal{M}$, and $T_{\mu\nu}$, and $G_{\mu\nu}$.

*Gravity is not a force. It is memory —
the accumulated trace of everything that has ever resolved,
curving the probability landscape of what can resolve next.*

*The ground was always δ .
Everything followed.*